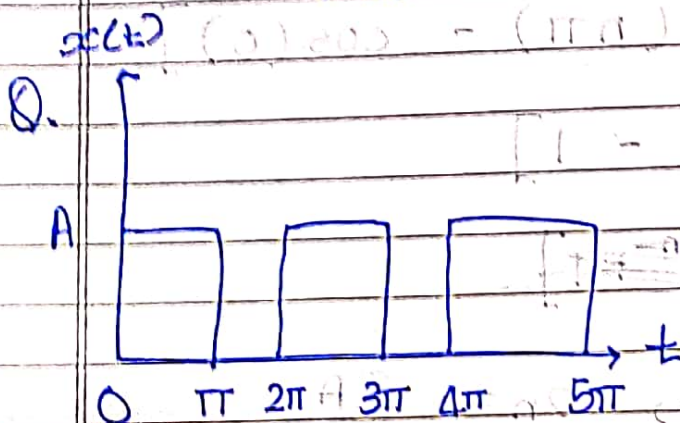


ISP

$$a_0 = \frac{1}{T} \int_0^T x(t) \cdot dt$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cdot \cos(n\omega t) \cdot dt$$

$$b_n = \frac{2}{T} \int_0^T x(t) \cdot \sin(n\omega t) \cdot dt$$



Here $T = 2\pi$

$$x(t) = A \quad 0 < t < \pi$$

$$x(t) = 0 \quad \pi < t < 2\pi$$

$$\omega = 2\pi f = 2\pi \frac{1}{T} = \frac{2\pi}{2\pi} = 1$$

① $a_0 = \frac{1}{T} \int_0^T x(t) \cdot dt$

$$= \frac{1}{T} \int_0^{2\pi} A \cdot dt + \int_{2\pi}^{4\pi} 0 \cdot dt$$

$$= \frac{1}{2\pi} [A \cdot t]_0^{2\pi} = \frac{A}{2}$$

② $a_n = \frac{2}{T} \int_0^T x(t) \cdot \cos(n\omega t) \cdot dt$

$$= \frac{2}{2\pi} \int_0^{\pi} A \cdot \cos(nt) \cdot dt + \int_{\pi}^{2\pi} 0 \cdot \cos(nt) \cdot dt$$

$$= \frac{A}{\pi} \left[\frac{\sin(nt)}{n} \right]_0^{\pi}$$

$$= \frac{A}{\pi n} (\sin(n\pi) - \sin(0))$$

$$= \frac{A}{\pi n} (0) = 0$$

$$b_n = \frac{2}{T} \int_0^T x(t) \cdot \sin(n\omega t) \cdot dt$$

$$= \frac{2}{2\pi} \int_0^{2\pi} (A) \cdot \sin(nt) \cdot dt$$

$$= \frac{A}{\pi} \left[-\frac{\sin(nt)}{n} \right]_0^{2\pi}$$

$$= \frac{A}{\pi} [\cos(n\pi) - \cos(0)]$$

$$= A - \frac{A}{\pi n} [(-1)^n - 1]$$

$$= \frac{A}{\pi n} [(-1)^n - 1]$$

if $n = \text{odd}$, $b_n = \frac{2A}{n\pi}$

if $n = \text{even}$, $b_n = 0$

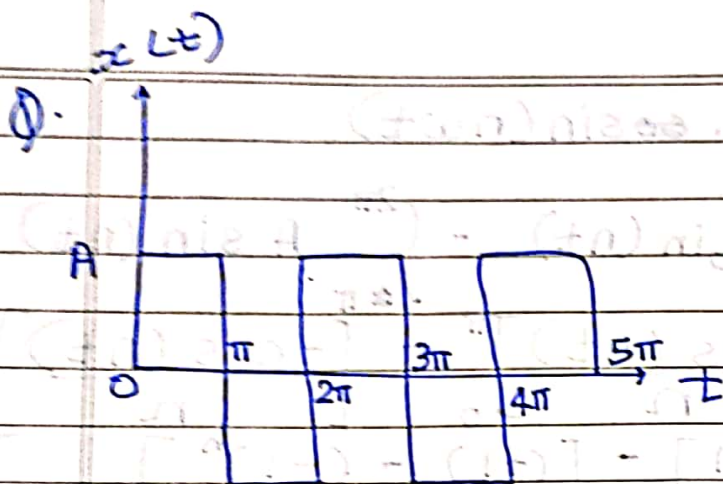
$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t))$$

$$= \frac{A}{2} + \sum_{n=1}^{\infty} (0 \cos(nt) +$$

$$= \frac{A}{2} + \sum_{n=1,3,5}^{\infty} \left[0 + \frac{2A}{\pi n} \right]$$

$$= \frac{A}{2} + \left[\frac{2A \sin(t)}{\pi} + \frac{2A \sin(3t)}{3\pi} + \right.$$

$$\left. \frac{2A \sin(5t)}{5\pi} + \dots \right]$$



$$\begin{aligned}\omega &= 2\pi f \\ &= 2\pi \frac{1}{2\pi} \\ &= 1\end{aligned}$$

$$x(t) = A \quad 0 < t < \pi$$

$$x(t) = -A \quad \pi < t < 2\pi$$

$$\begin{aligned}a_0 &= \frac{1}{T} \int_0^T x(t) \cdot dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} A \, dt - \int_{\pi}^{2\pi} A \, dt \\ &= \frac{1}{2\pi} [At]_0^{2\pi} - [At]_{\pi}^{2\pi} \\ &= \frac{1}{2\pi} [A\pi - 0] - [2A\pi - A\pi] \\ &= \frac{1}{2\pi} \{ A\pi - A\pi \} = 0\end{aligned}$$

$$\begin{aligned}a_n &= \frac{2}{T} \int_0^T x(t) \cdot \cos(n\omega t) \cdot dt \\ &= \frac{2}{2\pi} \int_0^{\pi} A \cos(nt) \cdot dt - \int_{\pi}^{2\pi} A \cos(nt) \\ &= \frac{A}{\pi} \left\{ \left[\frac{\sin(nt)}{n} \right]_0^{\pi} - \left[\frac{\sin(nt)}{n} \right]_{\pi}^{2\pi} \right\} \\ &= \frac{A}{\pi} [0 - 0] = 0\end{aligned}$$

$$b_n = \frac{2}{T} \int_0^T x(t) \cdot \sin(n\omega t) dt$$

$$= \frac{2}{2\pi} \int_0^{2\pi} A \sin(nt) - \int_{\pi}^{2\pi} A \sin(nt) dt$$

$$= \frac{A}{\pi} \left\{ \left[-\frac{\cos(nt)}{n} \right]_0^{\pi} - \left[-\frac{\cos(nt)}{n} \right]_{\pi}^{2\pi} \right\}$$

$$= -\frac{A}{n\pi} \left\{ [(-1)^n - 1] - [(+1) - (-1)^n] \right\}$$

$$= -\frac{A}{n\pi} [(-1)^n - 1 - 1 + (-1)^n]$$

$$= -\frac{A}{n\pi} [2(-1)^n - 2]$$

Here, when $n = 1, 3, 5$

$$b_n = \frac{4A}{n\pi}$$

when $n = 2, 4, 6$

$$b_n = 0$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega t) + b_n \sin(n\omega t)$$

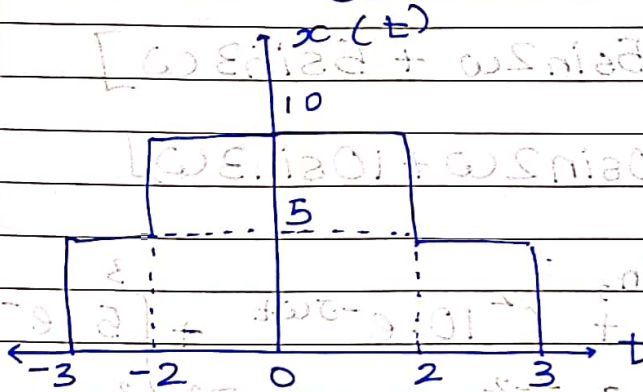
$$= 0 + \sum_{n=1,3,5}^{\infty} 0 + \frac{4A}{n\pi} \sin(nt)$$

$$= \left[\frac{4A}{\pi} \sin t + \frac{4A}{3\pi} \sin 3t + \dots \right]$$

$\therefore \text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0}{0 - 1} = -1$
 $\frac{e^{i\theta} - e^{-i\theta}}{2i}$ $\theta = \omega$
 $i = j$
 they are non-periodic $\therefore$ no 'T'

Fourier Transform $X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot dt$

Q.



Here,

$x(t) = 5 \quad -3 < t < -2$
 $= 10 \quad -2 < t < 2$
 $= 5 \quad 2 < t < 3$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot dt$$

$$= \int_{-3}^{-2} 5 \cdot e^{-j\omega t} \cdot dt + \int_{-2}^{2} 10 \cdot e^{-j\omega t} \cdot dt + \int_{2}^{3} 5 \cdot e^{-j\omega t} \cdot dt$$

$$= 5 \left[\frac{-e^{-j\omega t}}{j\omega} \right]_{-3}^{-2} + \left[\frac{10 e^{-j\omega t}}{-j\omega} \right]_{-2}^{2} + \left[\frac{5 e^{-j\omega t}}{-j\omega} \right]_{2}^{3}$$

$$= -5 \frac{e^{2j\omega} - e^{3j\omega}}{j\omega} - 10 \frac{e^{-2j\omega} - e^{2j\omega}}{j\omega} + 10 \frac{e^{2j\omega} - e^{3j\omega}}{j\omega}$$

$$= -5 \frac{e^{3j\omega} - e^{2j\omega}}{j\omega} + 5 \frac{e^{2j\omega} - e^{3j\omega}}{j\omega}$$

$$= -5 \frac{(e^{3j\omega} - e^{2j\omega}) - (e^{2j\omega} - e^{3j\omega})}{j\omega} + 10 \frac{(e^{2j\omega} - e^{-2j\omega})}{j\omega}$$

$$= 5 \frac{(e^{3j\omega} + e^{3j\omega} - e^{2j\omega} - e^{2j\omega})}{j\omega}$$

$$= \frac{1}{\omega} [10 \sin]$$

$$= 5 \left[\frac{-e^{j\omega 2}}{j\omega} + \frac{e^{j\omega 3}}{j\omega} \right] + 10 \left[\frac{-e^{-j\omega 2}}{j\omega} + \frac{e^{j\omega 2}}{j\omega} \right]$$

$$+ 5 \left[\frac{-e^{-j\omega 3}}{j\omega} + \frac{e^{-j\omega 2}}{j\omega} \right]$$

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$$\begin{aligned}
 &= \frac{2}{\omega} [10 \sin 2\omega + 5 \sin 3\omega - 5 \sin 2\omega] \\
 &= \frac{2}{\omega} [5 \sin 2\omega + 5 \sin 3\omega] \\
 &= \frac{1}{\omega} [10 \sin 2\omega + 10 \sin 3\omega]
 \end{aligned}$$

Solving Again,

$$\begin{aligned}
 &\int_{-3}^{-2} 5 e^{-j\omega t} + \int_{-2}^{-1} 10 e^{-j\omega t} + \int_{-1}^3 5 e^{-j\omega t} \\
 &5 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-3}^{-2} + 10 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-2}^{-1} + 5 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-1}^3 \\
 &5 \left[\frac{e^{j\omega 2}}{-j\omega} - \frac{e^{j\omega 3}}{-j\omega} \right] + 10 \left[\frac{e^{-j\omega 2}}{-j\omega} - \frac{e^{j\omega 2}}{-j\omega} \right] \\
 &+ 5 \left[\frac{e^{-j\omega 3}}{-j\omega} - \frac{e^{-j\omega 2}}{-j\omega} \right] \\
 &5 \left[\frac{e^{j\omega 3}}{j\omega} - \frac{e^{j\omega 2}}{j\omega} \right] + 10 \left[\frac{e^{j\omega 2}}{j\omega} - \frac{e^{-j\omega 2}}{j\omega} \right] + \\
 &5 \left[\frac{e^{-j\omega 2}}{j\omega} - \frac{e^{-j\omega 3}}{j\omega} \right]
 \end{aligned}$$

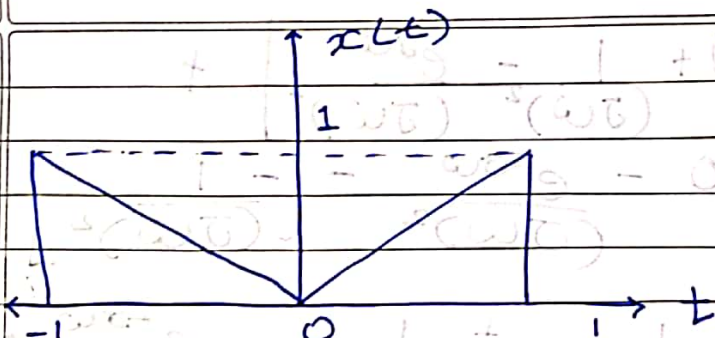
$$\frac{2}{\omega} [5 \sin 3\omega + 10 \sin 2\omega - 5 \sin 2\omega]$$

$$\frac{2}{\omega} [5 \sin 3\omega + 5 \sin 2\omega]$$

$$\frac{1}{\omega} [10 \sin 3\omega + 10 \sin 2\omega]$$

$$\int u \cdot v = u \int v - \int v \cdot u' \quad \text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0}{-1 - 0} = -1$$

Q.



$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$x(t) = -t \quad -1 < t < 0 \\ x(t) = t \quad 0 < t < 1$$

$$\begin{aligned} X(j\omega) &= \int_{-1}^0 -t \cdot e^{-j\omega t} dt + \int_0^1 t \cdot e^{-j\omega t} dt \\ &= \left[-\frac{t \cdot e^{-j\omega t}}{-j\omega} - \int \frac{e^{-j\omega t} \cdot (-1)}{-j\omega} \right]_{-1}^0 + \left[\frac{t \cdot e^{-j\omega t}}{-j\omega} - \int \frac{e^{-j\omega t}}{-j\omega} \right]_0^1 \\ &= \left[\frac{t \cdot e^{-j\omega t}}{j\omega} + \frac{e^{-j\omega t}}{j^2 \omega^2} \right]_{-1}^0 + \left[\frac{t \cdot e^{-j\omega t}}{-j\omega} - \frac{e^{-j\omega t}}{(j\omega)^2} \right]_0^1 \\ &= \left[0 - \left(\frac{-e^{j\omega}}{j\omega} \right) + \frac{1}{(j\omega)^2} - \frac{e^{j\omega}}{j^2 \omega^2} \right] + \left[-0 + \frac{e^{-j\omega}}{-j\omega} - \frac{e^{-j\omega}}{(j\omega)^2} \right] \\ &= \frac{1}{(j\omega)^2} - \frac{e^{j\omega}}{j\omega} + \frac{e^{j\omega}}{(j\omega)^2} - \frac{1}{(j\omega)^2} - \frac{e^{-j\omega}}{j\omega} + \frac{e^{-j\omega}}{(j\omega)^2} \\ &= \frac{e^{j\omega}}{j\omega} - \frac{e^{-j\omega}}{j\omega} + \frac{e^{j\omega}}{(j\omega)^2} - \frac{e^{-j\omega}}{(j\omega)^2} \\ &= \frac{e^{j\omega} - e^{-j\omega}}{j\omega} + \frac{e^{j\omega} - e^{-j\omega}}{(j\omega)^2} \\ &= \frac{2j \sin \omega}{j\omega} + \frac{2j \sin \omega}{(j\omega)^2} \\ &= \frac{2 \sin \omega}{\omega} - \frac{2 \sin \omega}{\omega^2} \end{aligned}$$

$$\left(\frac{e^{i\theta} + e^{-i\theta}}{2} \right) = \cos \theta \quad j^2 = -1 \quad \text{where } i = j$$

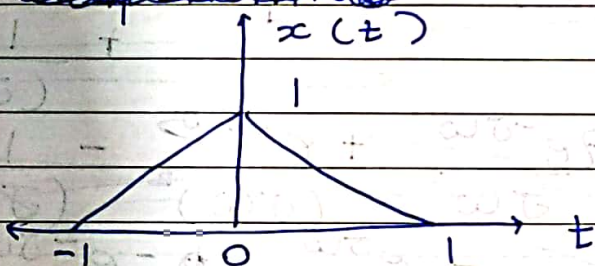
$$= \left[0 - \frac{(-1)e^{j\omega}}{j\omega} + \frac{1}{(j\omega)^2} - \frac{e^{j\omega}}{(j\omega)^2} \right] + \left[\frac{e^{-j\omega}}{-j\omega} - 0 - \frac{e^{-j\omega}}{(j\omega)^2} - \left(\frac{-1}{(j\omega)^2} \right) \right]$$

$$= \frac{e^{j\omega}}{j\omega} + \frac{1}{(j\omega)^2} + \frac{1}{(j\omega)^2} + \frac{e^{-j\omega}}{-j\omega} - \frac{e^{j\omega}}{(j\omega)^2} - \frac{e^{-j\omega}}{(j\omega)^2}$$

$$= \frac{e^{j\omega} - e^{-j\omega}}{j\omega} + \frac{2}{j^2 \omega^2} - \left(\frac{e^{j\omega} + e^{-j\omega}}{(j\omega)^2} \right)$$

$$= -\frac{2}{\omega^2} + \frac{2 \sin \omega}{\omega} + \frac{2 \cos \omega}{\omega^2}$$

Q. ~~Find the Fourier transform of the following signal~~



$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0}{-1 - 0} = -1$$

$$x(t) = 1 + t \quad -1 < t < 0$$

$$= 1 - t \quad 0 < t < 1$$

= 0 elsewhere.

$$X(j\omega) = \int_{-1}^0 (1+t) \cdot e^{-j\omega t} dt + \int_0^1 (1-t) e^{-j\omega t} dt$$

$$= \begin{bmatrix} e^{-\sigma\omega t} \\ -\sigma\omega \end{bmatrix}_0 - \begin{bmatrix} t \cdot e^{-\sigma\omega t} & e^{-\sigma\omega t} \\ -\sigma\omega & (-\sigma\omega)^2 \end{bmatrix}_1 + \begin{bmatrix} e^{-\sigma\omega t} \\ -\sigma\omega \end{bmatrix}_0$$

$$+ \begin{bmatrix} t \cdot e^{-\sigma\omega t} & e^{-\sigma\omega t} \\ -\sigma\omega & (-\sigma\omega)^2 \end{bmatrix}_0$$

$$= \frac{1}{-\sigma\omega} - \frac{e^{\sigma\omega}}{-\sigma\omega} - \left[\frac{-(1) \cdot e^{\sigma\omega}}{-\sigma\omega} - \frac{1}{(-\sigma\omega)^2} - \left(\frac{e^{\sigma\omega}}{(-\sigma\omega)^2} \right) \right]$$

$$+ \frac{e^{\sigma\omega}}{-\sigma\omega} - \frac{1}{-\sigma\omega} + \left[\frac{e^{\sigma\omega}}{(-\sigma\omega)^2} + \frac{1 \cdot e^{\sigma\omega}}{(-\sigma\omega)^2} \right]$$

$$= -\frac{1}{\sigma\omega} + \frac{e^{\sigma\omega}}{\sigma\omega} + \frac{e^{\sigma\omega}}{\sigma\omega} + \frac{1}{(\sigma\omega)^2} - \frac{e^{\sigma\omega}}{(\sigma\omega)^2}$$

$$= -\frac{1}{\sigma\omega} + \frac{e^{\sigma\omega}}{\sigma\omega} + \frac{1}{(\sigma\omega)^2} - \frac{e^{\sigma\omega}}{(\sigma\omega)^2}$$

$$= \frac{2e^{\sigma\omega}}{\sigma\omega}$$

$$P = \frac{e^{\sigma\omega}}{\sigma\omega} - \left(\frac{-1}{\sigma\omega^2} \cdot \frac{e^{\sigma\omega}}{\sigma\omega} \right)$$

$$= \begin{bmatrix} e^{-\sigma\omega(0)} & e^{-\sigma\omega(-1)} \\ -\sigma\omega & -\sigma\omega \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ \sigma^2\omega^2 & -\sigma\omega \end{bmatrix} - \frac{(-1)e^{-\sigma\omega(-1)}}{\sigma^2\omega^2}$$

$$+ \begin{bmatrix} e^{-\sigma\omega} & -1 \\ -\sigma\omega & -\sigma\omega \end{bmatrix} + \begin{bmatrix} e^{-\sigma\omega} & e^{-\sigma\omega} \\ \sigma^2\omega^2 & \sigma^2\omega^2 \end{bmatrix} - \frac{1}{\sigma^2\omega^2}$$

$$= -\frac{1}{\sigma\omega} + \frac{e^{\sigma\omega}}{\sigma\omega} + \frac{1}{\sigma^2\omega^2} + \frac{e^{\sigma\omega}}{\sigma^2\omega^2} + \frac{e^{\sigma\omega}}{\sigma^2\omega^2} - \frac{1}{\sigma\omega}$$

$$+ \frac{1}{\sigma\omega} - \frac{e^{-\sigma\omega}}{\sigma\omega} - \frac{e^{-\sigma\omega}}{\sigma^2\omega^2} - \frac{1}{\sigma\omega^2}$$

$$= \frac{2e^{\sigma\omega}}{\sigma\omega} + \frac{e^{\sigma\omega}}{\sigma^2\omega^2} - \frac{e^{-\sigma\omega}}{\sigma^2\omega^2}$$

$$= \frac{2e^{\sigma\omega}}{\sigma\omega} - \frac{e^{\sigma\omega} + e^{-\sigma\omega}}{\omega^2}$$

$$= \left[\frac{(1-t) \cdot e^{-j\omega t}}{-j\omega} - (-1) \frac{e^{-j\omega t}}{(j\omega)^2} \right]_0^{-1}$$

$$= \left[\frac{(1+t) \cdot e^{-j\omega t}}{-j\omega} - \frac{e^{-j\omega t}}{(j\omega)^2} \right]_0^{-1}$$

$$= \left[\frac{2e^{j\omega}}{-j\omega} - \frac{e^{j\omega}}{(j\omega)^2} \right] - \left[\frac{2e^{-j\omega}}{-j\omega} - \frac{e^{-j\omega}}{(j\omega)^2} \right]$$

$$= \left[\frac{-2e^{j\omega}}{-j\omega} - \frac{e^{j\omega}}{(j\omega)^2} \right] + \left[\frac{2e^{-j\omega}}{-j\omega} + \frac{e^{-j\omega}}{(j\omega)^2} \right]$$

$$= \left[\frac{2e^{j\omega}}{j\omega} - \frac{e^{j\omega}}{(j\omega)^2} \right] + \left[\frac{-2e^{-j\omega}}{j\omega} - \frac{e^{-j\omega}}{(j\omega)^2} \right]$$

$$= \left[\frac{2e^{j\omega}}{j\omega} - \frac{e^{j\omega}}{(j\omega)^2} \right] + \left[\frac{-2e^{-j\omega}}{j\omega} - \frac{e^{-j\omega}}{(j\omega)^2} \right] + \left[\frac{1}{j\omega} + \frac{e^{-j\omega}}{j\omega} \right]$$

$$= \frac{2}{\omega} \left(\frac{e^{j\omega} - e^{-j\omega}}{j} \right) + \frac{1}{\omega} \left(\frac{e^{j\omega} + e^{-j\omega}}{j} \right)$$

$$= \left[\left(\frac{1}{-j\omega} + \frac{1}{j^2\omega^2} \right) - \left(\frac{-2e^{j\omega}}{-j\omega} + \frac{e^{j\omega}}{j^2\omega^2} \right) \right] +$$

$$\left[\left(\frac{2e^{-j\omega}}{-j\omega} - \frac{e^{-j\omega}}{j^2\omega^2} \right) - \left(\frac{1}{-j\omega} - \frac{1}{j^2\omega^2} \right) \right]$$

$$= \left[\frac{-1}{j\omega} + \frac{1}{j^2\omega^2} - \frac{2e^{j\omega}}{j\omega} + \frac{e^{j\omega}}{j^2\omega^2} - \frac{2e^{-j\omega}}{j\omega} + \frac{e^{-j\omega}}{j^2\omega^2} + \frac{1}{j\omega} + \frac{1}{j^2\omega^2} \right]$$

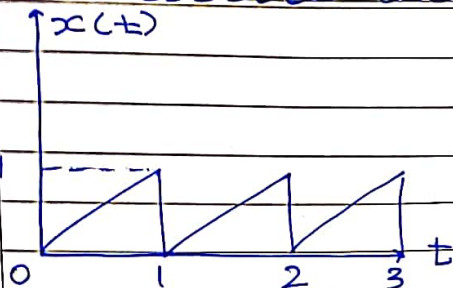
$$= \frac{2}{j^2\omega^2} + \frac{2\cos \omega}{\omega^2}$$

ISP

$$e^{-jn\pi} = (-1)^n = \cos n\pi - j\sin n\pi \quad \left\{ \begin{aligned} e^{-jn2\pi} &= \cos 2\pi - j\sin 2\pi = 1 \\ e^{-jn\pi} &= \cos n\pi - j\sin n\pi \end{aligned} \right.$$

Exponential Form of Fourier

Q.1)



$$x(t) = t \quad 0 < t < 1$$

$$T = 1$$

$$\omega = 2\pi$$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n \cdot e^{jn\omega t}$$

$$\text{Where } C_n = \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} dt$$

$$\therefore C_n = \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} dt$$

$$= \frac{1}{1} \int_0^1 t \cdot e^{-jn\omega t} dt$$

$$= \left[\frac{t \cdot e^{-jn\omega t}}{-jn\omega} - \frac{e^{-jn\omega t}}{(jn\omega)^2} \right]_0^1$$

$$= \left[\frac{e^{-jn\omega}}{-jn\omega} - \frac{e^{-jn\omega}}{(jn\omega)^2} + \frac{1}{(jn\omega)^2} \right]$$

$$= \frac{-1}{jn2\pi} \left[1 + \frac{1}{jn2\pi} + \frac{1}{jn2\pi} \right]$$

$$= \frac{-1}{jn2\pi}$$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n \cdot e^{jn\omega t}$$

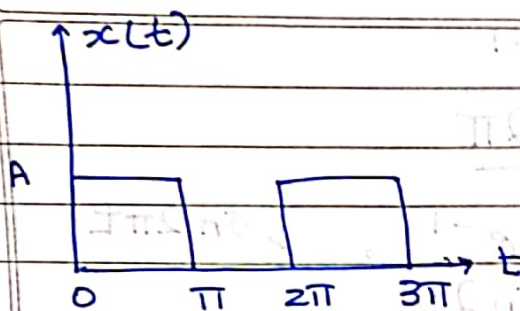
$$= \sum_{n=-\infty}^{\infty} \frac{-1}{jn2\pi} \cdot e^{jn2\pi t}$$

time period if

$$A e^{-at}$$

ssssssssssssssss

Q.2



$$T = 2\pi$$

$$\omega = 1 = 2\pi f$$

$$x(t) = A \quad 0 < t < \pi$$

$$= 0 \quad \pi < t < 2\pi$$

$$C_n = \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} dt$$

$$= \frac{1}{2\pi} \int_0^{\pi} A \cdot e^{-jn t} dt$$

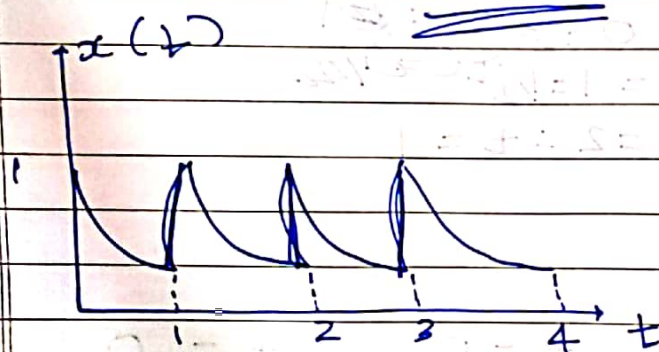
$$= \frac{A}{2\pi} \left[\frac{e^{-jn t}}{-jn} \right]_0^{\pi} = \frac{A}{2\pi} \left[\frac{e^{-jn\pi}}{-jn} - \frac{1}{-jn} \right]$$

$$= \frac{A}{2\pi jn} (-(-1)^n + 1)$$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n \cdot e^{jn\omega t}$$

$$= \sum_{n=-\infty}^{\infty} -\frac{A}{2\pi jn} ((-1)^n - 1) \cdot e^{jnt}$$

Q.



$$T = 1$$

$$\omega = 2\pi$$

$$x(t) = e^{-t} \quad 0 < t < 1$$

$$C_n = \frac{1}{T} \int_0^T e^{-t} \cdot e^{-jn2\pi t} dt$$

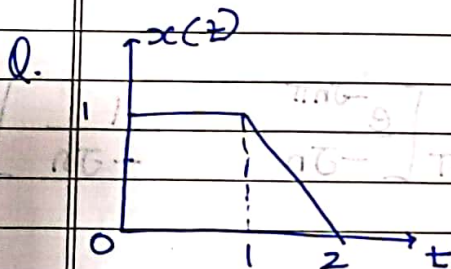
$$= \left[\frac{e^{-t(1+jn2\pi)}}{-(1+jn2\pi)} \right]_0^1 = \frac{e^{-1-jn2\pi}}{-1-jn2\pi} - \frac{1}{-1-jn2\pi}$$

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$$= \frac{1 - e^{-1}}{1 + jn2\pi}$$

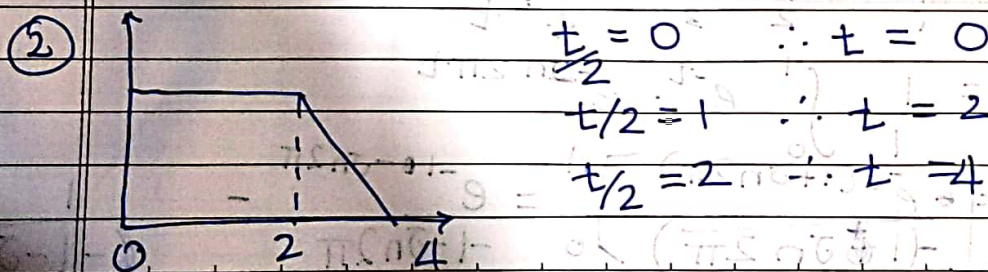
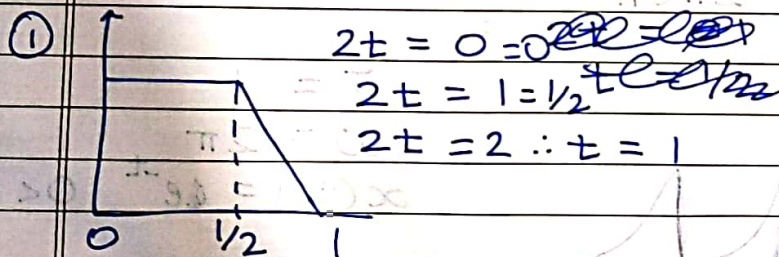
$$x(t) = \sum_{-\infty}^{\infty} \frac{1 - e^{-1}}{1 + jn2\pi} \cdot e^{jn2\pi t}$$

* ISP m1

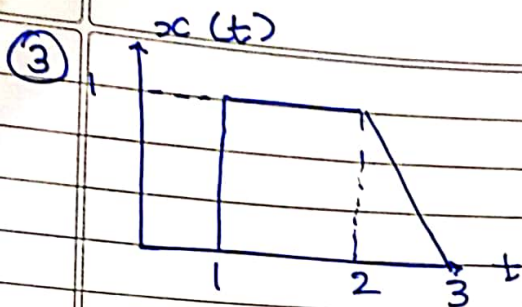


- Draw
- ① $x(2t)$
 - ② $x(t/2)$
 - ③ $x(t-1)$
 - ④ $x(t+1)$
 - ⑤ $x(1-t)$
 - ⑥ $x(-1-t)$

Order → ① Folding ② Scaling ③ Shifting
④ Amplitude



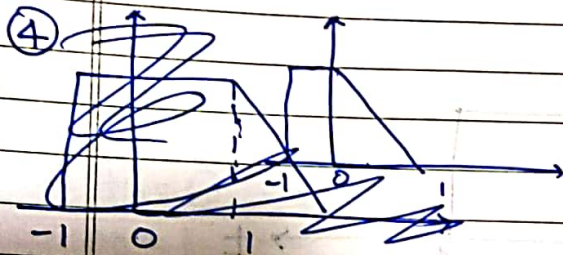
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$$t-1=0 \therefore t=1$$

$$t-1=1 \therefore t=2$$

$$t-1=2 \therefore t=3$$



$$t+1=0 \therefore t=-1$$

$$t+1=1 \therefore t=0$$

$$t+1=2 \therefore t=1$$

⑤

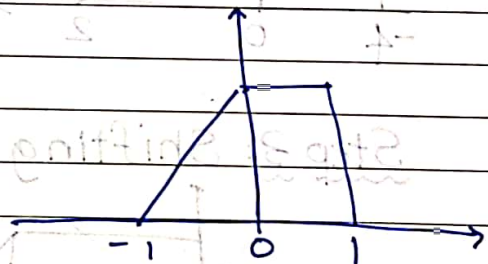
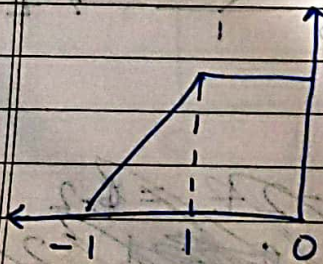
$$1-t=0 \therefore t=1$$

$$1-t=1 \therefore t=0$$

$$1-t=2 \therefore t=-1$$

Step 1: Folding

Step 2: Shifting

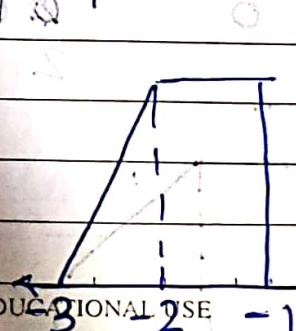
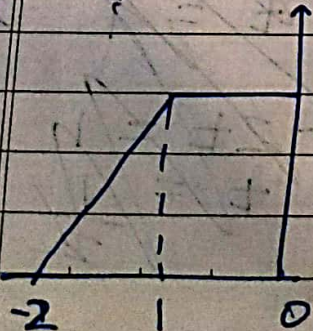


⑥

$$x(-1-t)$$

Step 1: Folding

Step 2: Shifting

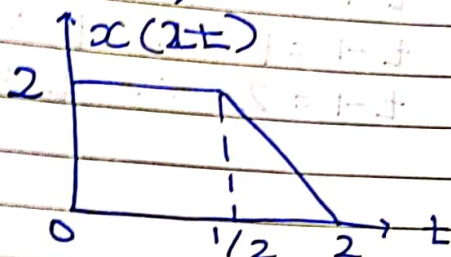


* Q.

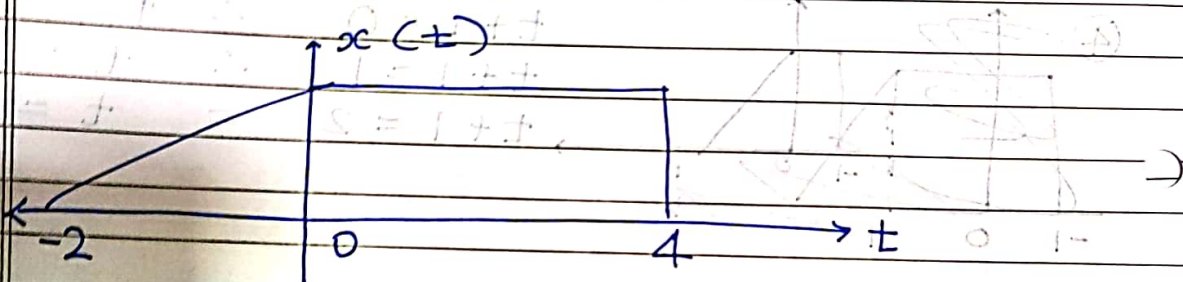
amplitude scaling

$$2x(2t)$$

2(1)
↓
original
amplitude

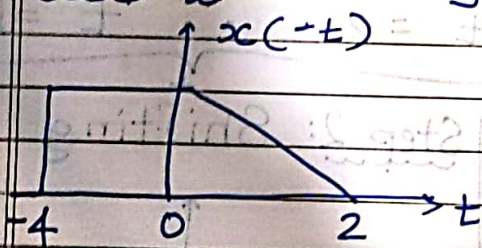


Q.

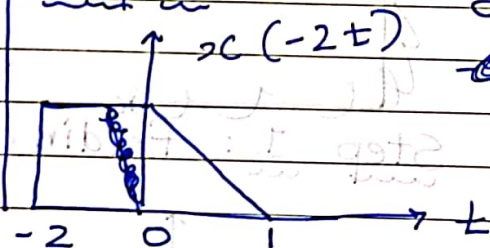


① $x(2-2t)$

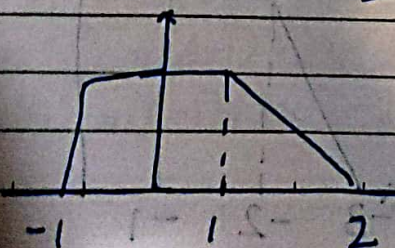
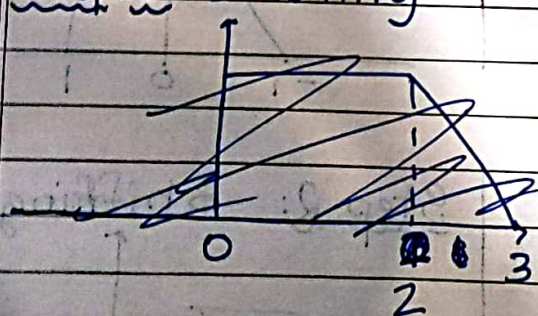
Step 1: Folding



Step 2: Scaling



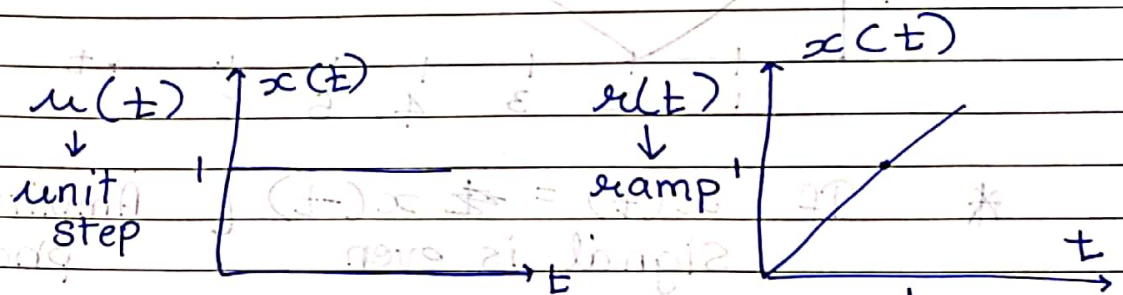
Step 3: Shifting



$$\begin{aligned} 2-2t &= t-2 \\ 2t &= 0 \\ t &= 0 \\ 2-2t &= 0 \\ t &= 1 \\ 2-2t &= 0 \\ t &= 1 \end{aligned}$$

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Q. $x(t) = u(t) + r_1(t-1) - 2r_1(t-2) + r_1(t-3) - u(t-4)$



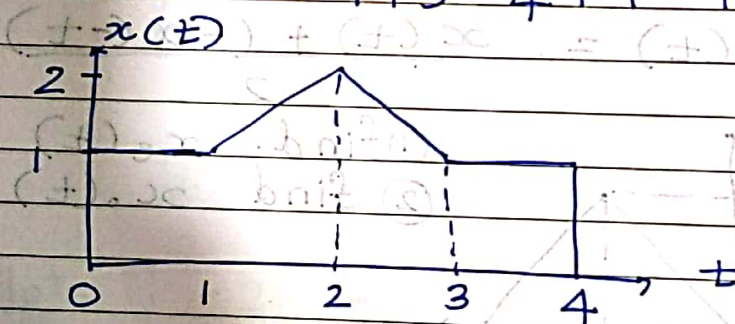
$t=0 \rightarrow u(0) = 1$

$t=1 \rightarrow u(1) + r_1(1-1) = 1$

$t=2 \rightarrow u(2) + r_1(2-1) - 2r_1(2-2) = 1 + 1 - 0 = 2$

$t=3 \rightarrow u(3) + r_1(3-1) - 2r_1(3-2) + r_1(3-3) = 1 + 2 - 2 + 0 = 1$

$t=4 \rightarrow u(4) + r_1(4-1) - 2r_1(4-2) + r_1(4-3) - u(4-4)$
 $= 1 + 3 - 4 + 1 - 1 = 0$



Q. $\tilde{x}(t) = u(t) - r_1(t-1) + 2r_1(t-2) - r_1(t-3) + u(t-4) - 2u(t-5)$

$t=0 \rightarrow u(0) = 1$

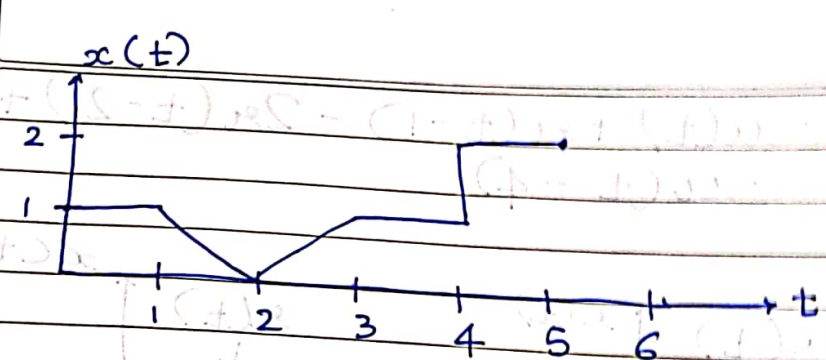
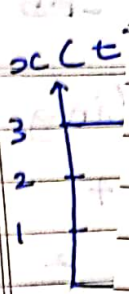
$t=1 \rightarrow u(1) - r_1(1-1) = 1 - 0 = 1$

$t=2 \rightarrow u(2) - r_1(2-1) + 2r_1(2-2) = 1 - 1 + 0 = 0$

$t=3 \rightarrow u(3) - r_1(3-1) + 2r_1(3-2) - r_1(3-3) = 1 - 2 + 2 - 0 = 1$

$t=4 \rightarrow u(4) - r_1(4-1) + 2r_1(4-2) - r_1(4-3) + u(4-4) = 1 - 3 + 4 - 1 + 1 = 2$

$t=5 \rightarrow u(5) - r_1(5-1) + 2r_1(5-2) - r_1(5-3) + u(5-4) - 2u(5-5)$
 $= 1 - 4 + 6 - 2 + 1 - 2 = 0$

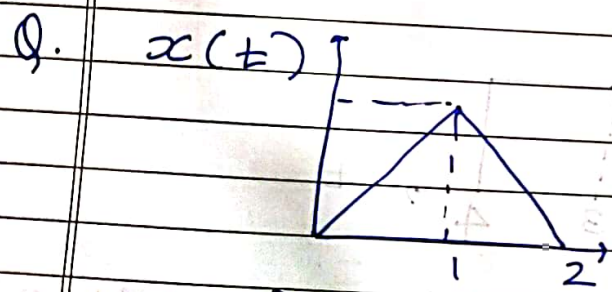


- ① * If $x(t) = x(-t)$ } Amplitude is equal
Signal is even } phase is same
- $x(t) = -x(-t)$ } Phase is opp
Signal is odd } but Amplitude is same

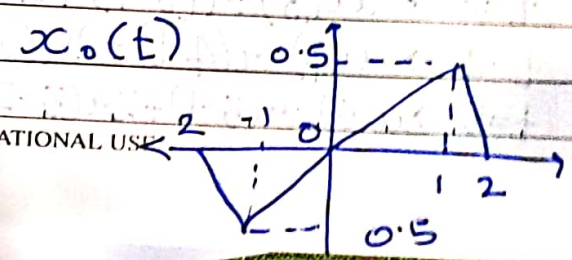
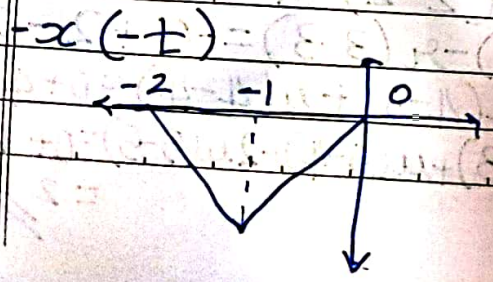
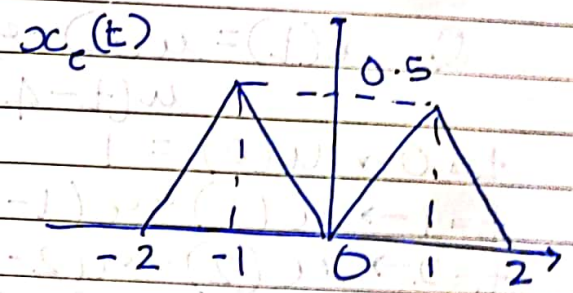
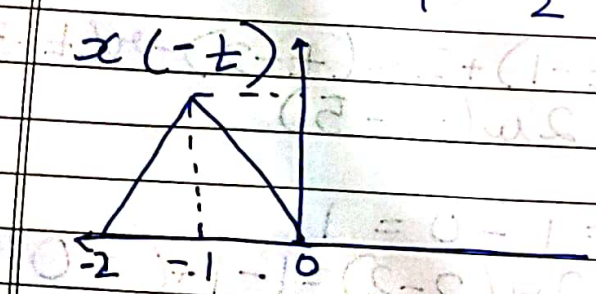
② *

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

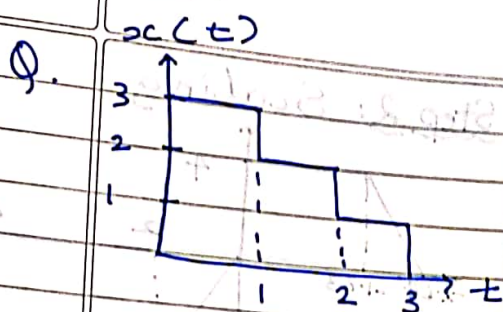
$$x_o(t) = \frac{x(t) - x(-t)}{2}$$



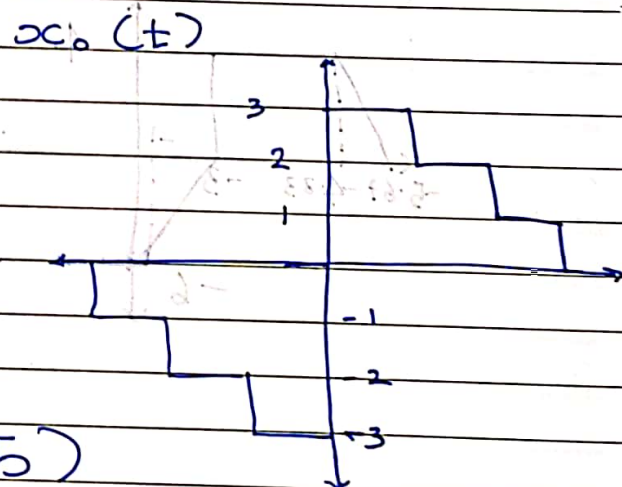
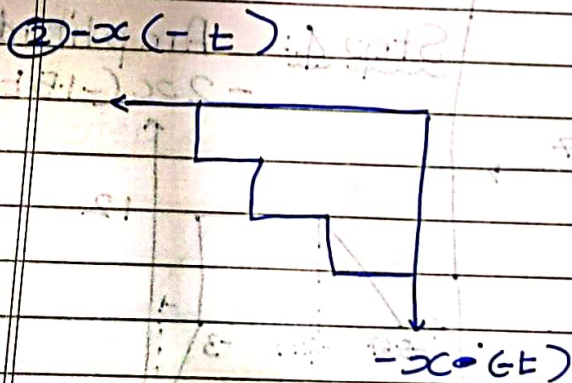
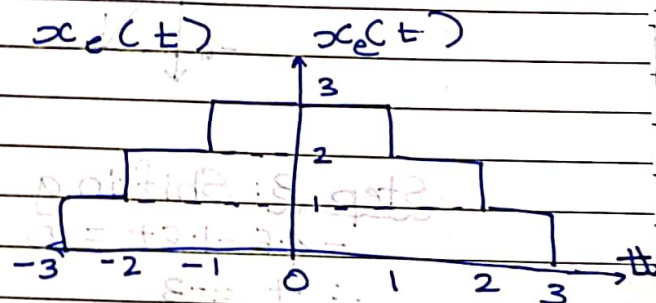
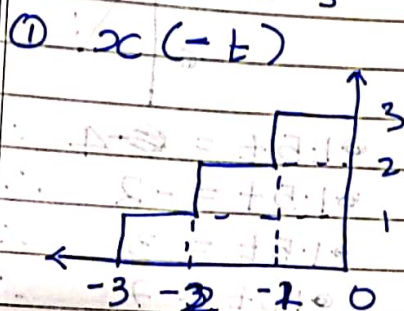
- ① find $x_e(t)$
② find $x_o(t)$



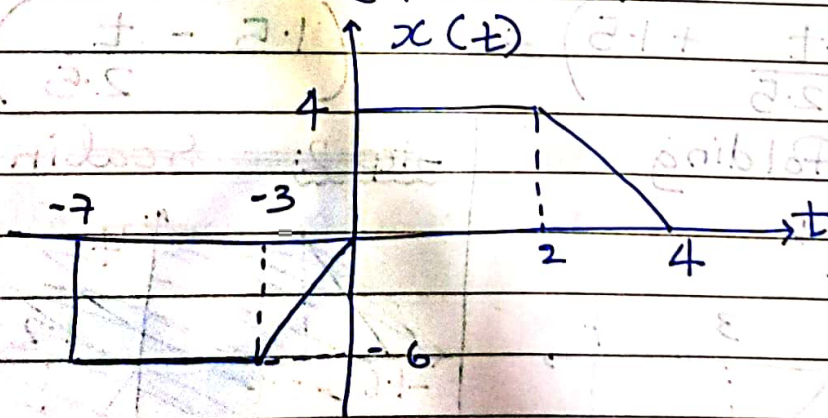
FOR EDUCATIONAL USE



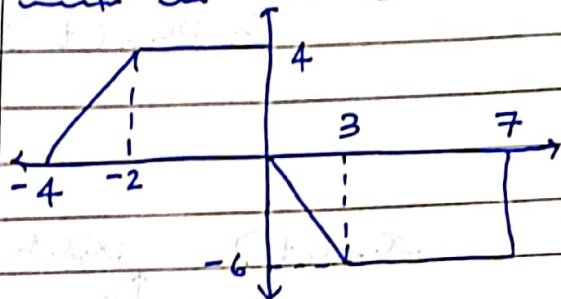
Construct ① $x_e(t)$
② $x_o(t)$



Q. Plot $y = -2x(-1.5t - 4.5)$
 $= -2x(-4.5 - 1.5t)$



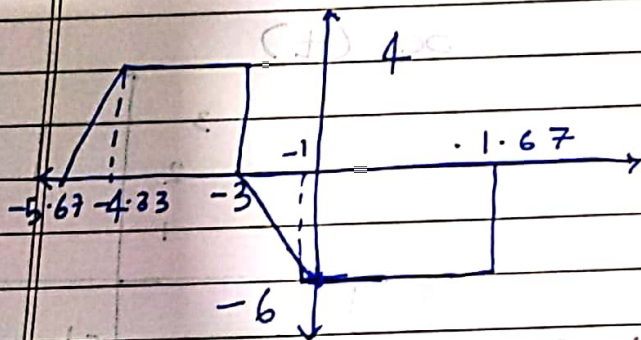
Step 1: folding



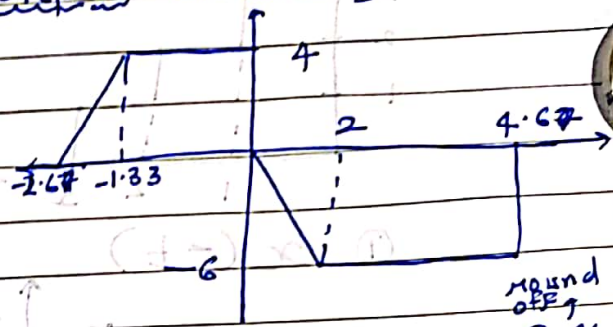
Step 3: Shifting

$$-4.5 - 1.5t = 0$$

$$\therefore t = -3$$



Step 2: Scaling



$$\bullet 1.5t = -4 \therefore t = -2.66$$

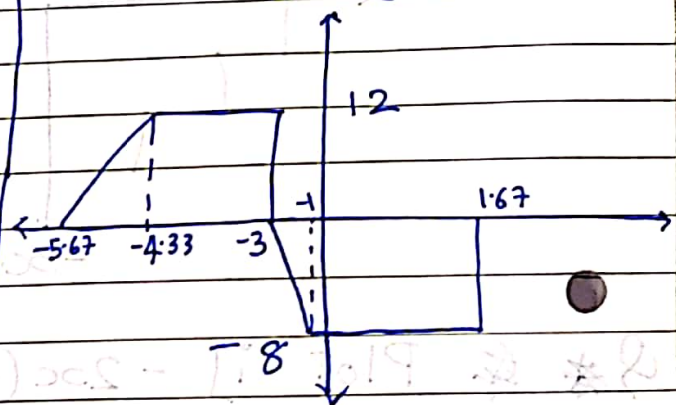
$$\bullet 1.5t = -2 \therefore t = -1.33$$

$$\bullet 1.5t = 3 \therefore t = 2$$

$$\bullet 1.5t = 7 \therefore t = 4.66$$

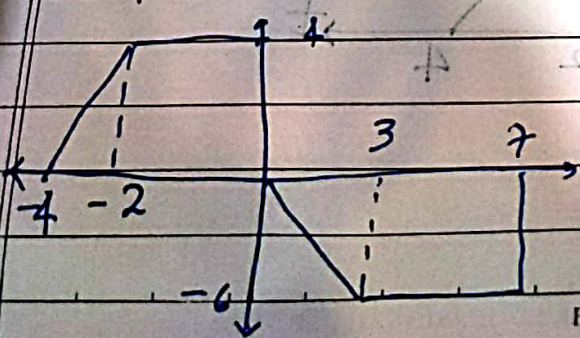
Step 4: Amplitude

$$-200(-1.5t - 4.5)$$

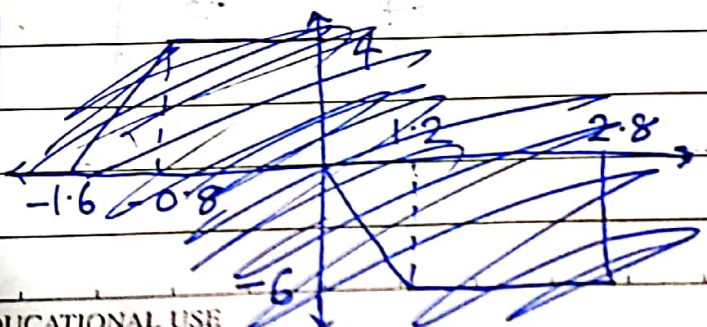


$$ii) \quad 2x \left(\frac{-t}{2.5} + 1.5 \right) = 200 \left(1.5 - \frac{t}{2.5} \right)$$

Step 1: Folding



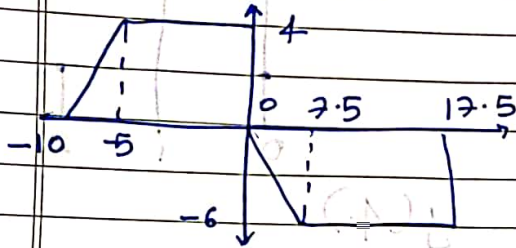
Step 2: Scaling



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$$\frac{t}{2.5} = 7 \Rightarrow t = 17.5$$

Step 2: Scaling:

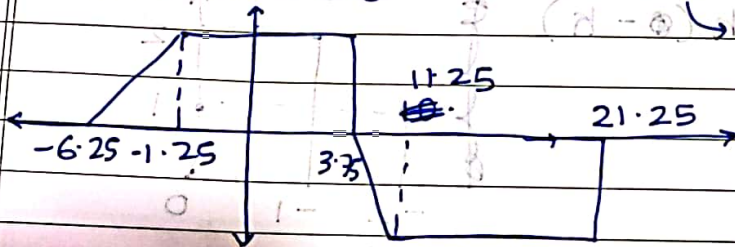


$$\frac{t}{2.5} = -2 \Rightarrow t = -5$$

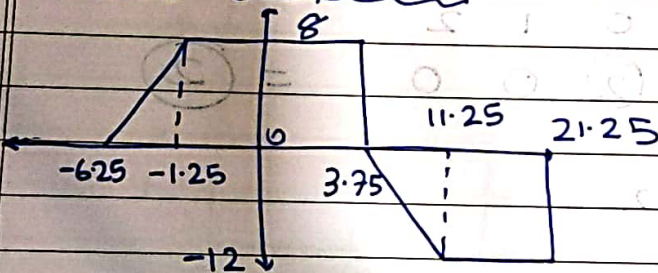
$$\frac{t}{2.5} = -4 \Rightarrow t = -10$$

$$\frac{t}{2.5} = 3 \Rightarrow t = 7.5$$

Step 3: Shifting: $(1.5 - t/2.5) = 0$



Step 4: Amplitude



$$200 \left(\frac{-t}{2.5} + 1.5 \right)$$

* Convolution Signals (a.k.a. Discrete Signal)

You will be given $x(k)$ & $h(k)$ signal graphs. Find

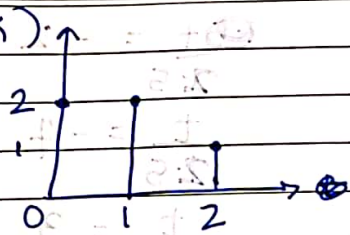
$$y(n) = \sum_{-\infty}^{\infty} x(k) \cdot h(n-k)$$

for all $n = l + m - 1$

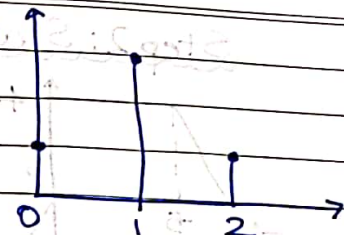
no. samples in x no. of samples in y

Q

$x(k)$



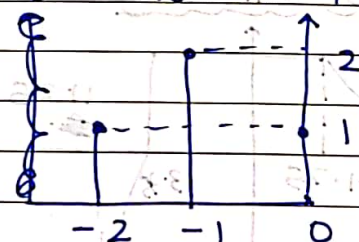
$h(k)$



$y(0)$ to $y(4)$

no. of convolutions = $l+m-1 = 3+3-1 = 5$

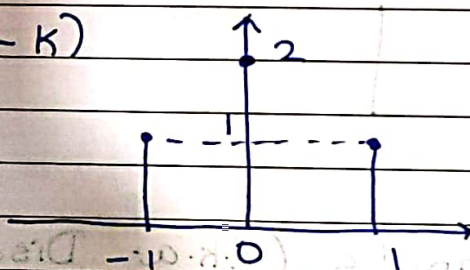
→ Step 1: $h(0-k)$



$$y(0) = \sum x(k) \cdot h(-k)$$

$$\begin{matrix} -2 & -1 & 0 & 1 & 2 \\ \sum & 0 & 0 & 2 & 0 & 0 \end{matrix} = 2$$

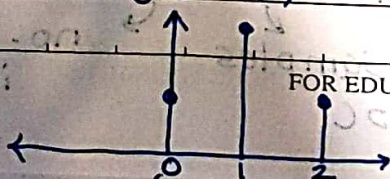
→ $h(1-k)$



$$y(1) = \sum x(k) \cdot h(1-k)$$

$$\begin{matrix} -2 & -1 & 0 & 1 & 2 \\ \sum & 0 & 4 & 2 & 1 & 0 \end{matrix} = 6$$

→ $y(2) = \sum x(k) \cdot h(2-k)$

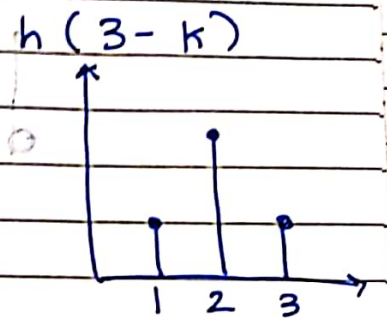


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$$\textcircled{2} + \textcircled{4} + \textcircled{1} = \textcircled{7}$$

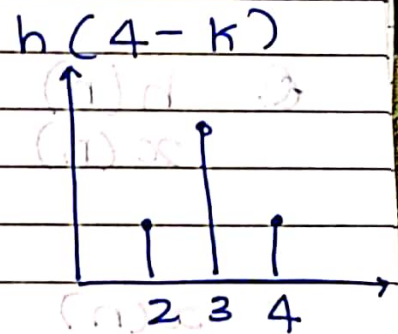
$$\rightarrow y(3) = x(k) \cdot h(3-k)$$

$$\begin{array}{cccc} 0 & 1 & 2 & 3 \\ 0 & 0 & 2 & 2 & 0 \end{array} = \textcircled{4}$$

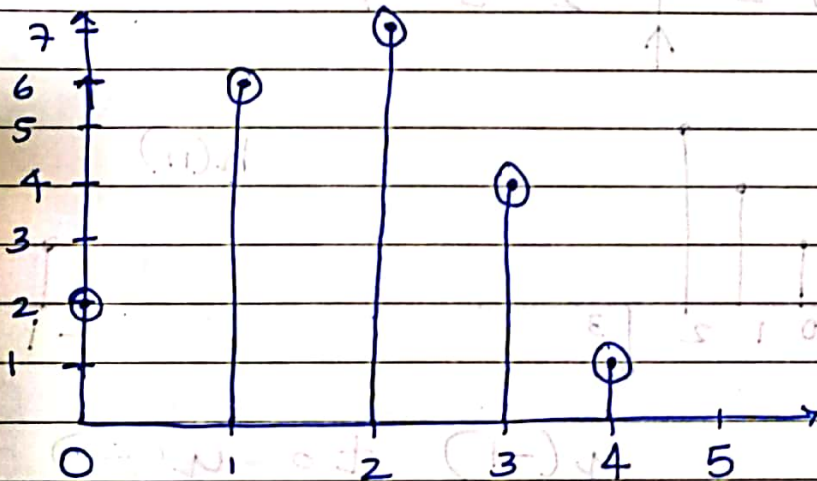


$$\rightarrow y(4) = x(k) \cdot h(4-k)$$

$$\begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 0 & 0 \end{array} = \textcircled{1}$$



Answer:

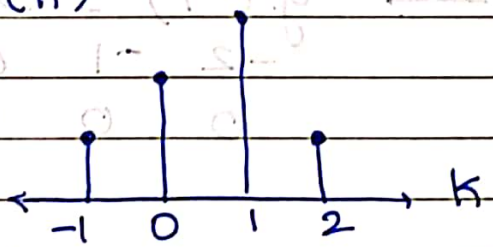
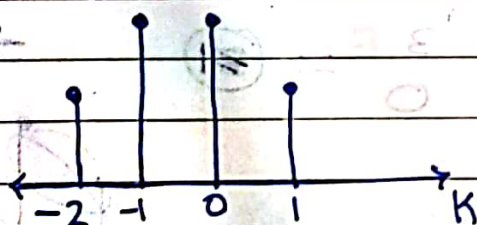


Solve

Q.

$$x(k)$$

$$h(k)$$



Q]

$$h(n) = 1 \quad 2 \quad 1 \quad -1$$

$$x(n) = 1 \quad 2 \quad 3 \quad -1$$

$$x(n) \uparrow$$

$h(n)$



~~$y(0)$ to $y($~~

Q.

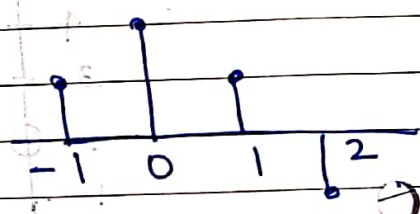
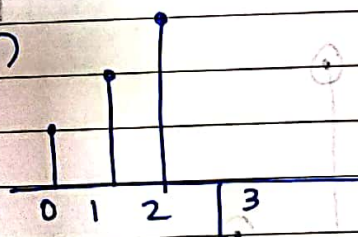
$$h(n) = 1 \quad 2 \quad 1 \quad -1$$

$$x(n) = 1 \quad 2 \quad 3 \quad -1$$

$\uparrow$

$x(n)$

$h(n)$



$y(-1)$ to $y(5)$

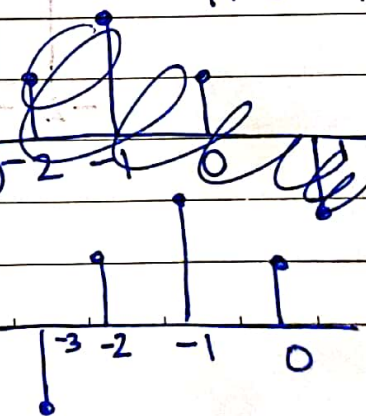
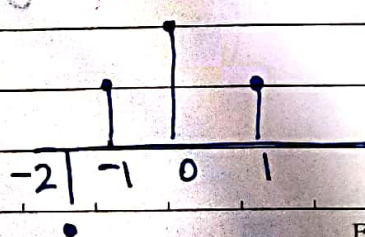
$$\rightarrow y(-1) = \sum_k x(k) h(-1-k)$$

$$\begin{matrix} -2 & -1 & 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{matrix} = \textcircled{1}$$

$$\begin{matrix} h(-k) \\ -1-k=0 \\ k=-1 \end{matrix}$$

$h(-k)$

$h(-1-k)$



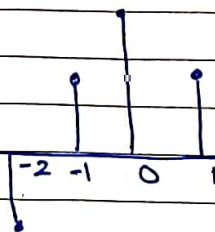
FOR EDUCATIONAL USE

$$\rightarrow y(0) = x(k) \cdot h(-k)$$

-2	-1	0	1	2
0	0	2	2	0

 $= \textcircled{4}$

$h(-k)$



$$\rightarrow y(1) = x(k) \cdot h(1-k)$$

-1	0	1	2	3
0	1	4	3	0

 $= \textcircled{8}$

$h(1-k)$

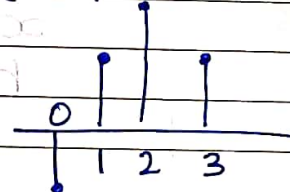


$$\rightarrow y(2) = x(k) \cdot h(2-k)$$

0	1	2	3
-1	2	6	-1

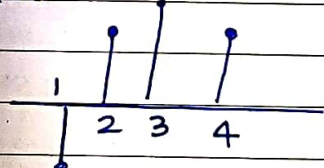
 $= \textcircled{6}$

$h(2-k)$



$$\rightarrow y(3) = x(k) \cdot h(3-k)$$

$h(3-k)$



0	1	2	3	4
0	2	6	2	0

 $= \textcircled{10}$

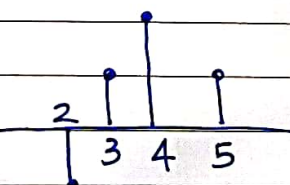
0	1	2	3	4
0	-1	2	-2	0

 $= \textcircled{-1}$

$$\rightarrow y(4) = x(k) \cdot h(4-k)$$

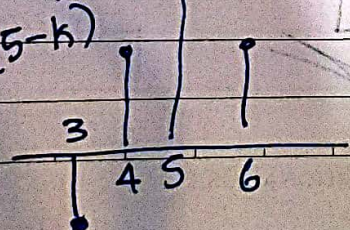
0	1	2	3	4	5
0	0	-3	-1	0	0

 $= \textcircled{-4}$



$$\rightarrow y(5) = x(k) \cdot h(5-k)$$

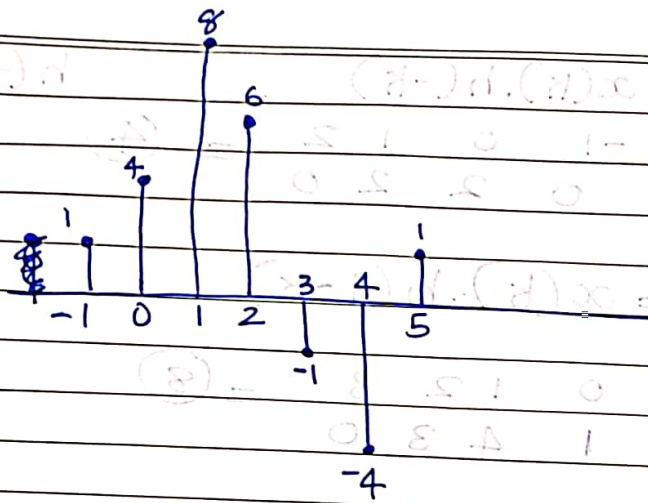
$h(5-k)$



0	1	2	3	4	5	6
0	0	0	1	0	0	0

FOR EDUCATIONAL USE

$= \textcircled{1}$



* Continuous - Convolution:

$x(k) \rightarrow$ lower limit

$h(k) \rightarrow$ upper limit

Q. $x(t) = e^{-at} \cdot u(t)$
 $h(t) = u(t)$

$$x(k) = e^{-ak} \cdot u(k)$$

$$h(t-k) = u(t-k)$$

$$t-k=0$$

$$k=t$$

$$\begin{aligned} \therefore y(t) &= \int_{-\infty}^{\infty} x(k) \cdot h(t-k) dk = \int_{-\infty}^{\infty} e^{-ak} \cdot u(k) \cdot u(t-k) dk \\ &= \int_0^t e^{-ak} dk = \left[\frac{e^{-ak}}{-a} \right]_0^t = \frac{e^{-at} - 1}{-a} \\ &= \frac{1 - e^{-at}}{a} \end{aligned}$$

Q. $x(t) = e^{-t} \cdot u(t)$
 $h(t) = e^{-2t} \cdot u(t)$

$\therefore y(t) = \int_{-\infty}^{\infty} x(k) \cdot h(t-k) dk$

$\therefore y(t) = \int_{-\infty}^{\infty} e^{-k} \cdot u(k) \cdot e^{-2(t-k)} \cdot u(t-k) dk$
 $= \int_{-\infty}^{\infty} e^{-k} \cdot e^{-2(t-k)} dk$
 $= \int_{-\infty}^{\infty} e^{-2t+k} dk = \int_{-\infty}^{\infty} e^{-2t} \cdot e^k dk$
 $= e^{-2t} \int_{-\infty}^{\infty} e^k dk = e^{-2t} [e^k]_0^t$
 $= e^{-2t} (e^t - 1)$

Q. $x(t) = e^{-4t} \cdot u(t)$
 $h(t) = u(t-4)$

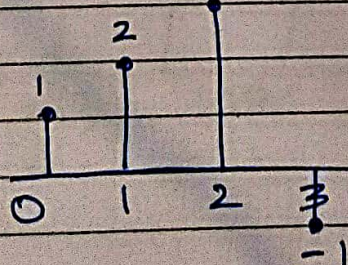
$y(t) = \int_{-\infty}^{\infty} x(k) \cdot h(t-k) dk$

$t-k-4$
 $t-4 = k$
 $\nearrow$ limits

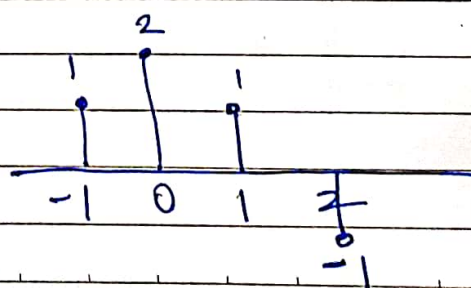
$y(t) = \int_{-\infty}^{\infty} e^{-4k} \cdot u(k) \cdot u(t-k-4) dk$
 $= \left[\frac{e^{-4k}}{-4} \right]_0^{t-4} = \frac{e^{-4(t-4)}}{-4} \cdot \frac{1}{-4}$
 $= \frac{1 - e^{-4(t-4)}}{4}$

Solve

Q. $x(n)$



$h(n)$



Con

$$\boxed{e^{-\infty} = 0}$$

Q. $x(t) = e^t$
 $h(t) = (3e^{-2t} - 1) \cdot u(t)$

$$y(t) = \int_{-\infty}^{\infty} x(k) \cdot h(t-k) dk$$

$$\therefore y = \int_{-\infty}^{\infty} e^k \cdot 3e^{-2(t-k)} \cdot u(t-k) dk$$

$$= 3 \int_{-\infty}^t e^k \cdot e^{-2t+2k} dk$$

$$= 3e^{-2t} \int_{-\infty}^t e^{3k} dk$$

$$= 3e^{-2t} \left[\frac{e^{3k}}{3} \right]_{-\infty}^t$$

$$= 3e^{-2t} (e^{3t} - 1)$$

$$\rightarrow = \int_{-\infty}^{\infty} e^k \cdot (3e^{-2(t-k)} - 1) dk$$

$$= \int_{-\infty}^t e^k \cdot 3 \cdot e^{-2t+2k} dk - \int_{-\infty}^t e^k dk$$

$$= 3e^{-2t} \int_{-\infty}^t e^{3k} dk - \int_{-\infty}^t e^k dk$$

$$= 3e^{-2t} \left[\frac{e^{3k}}{3} \right]_{-\infty}^t - \left[e^k \right]_{-\infty}^t$$

$$= e^{3t} - e^t$$

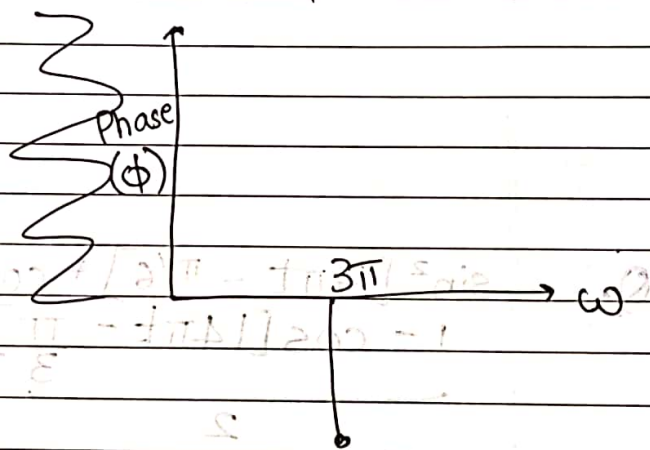
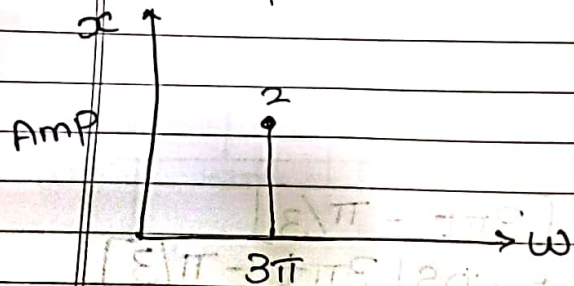
$$= 0$$

* Single & Double side representation
 Compare with $a \cos(\omega t - \theta)$
 ↳ phase

Q. Draw Amplitude & phase
 $x(t) = 2 \cos(3\pi t - \frac{\pi}{4})$

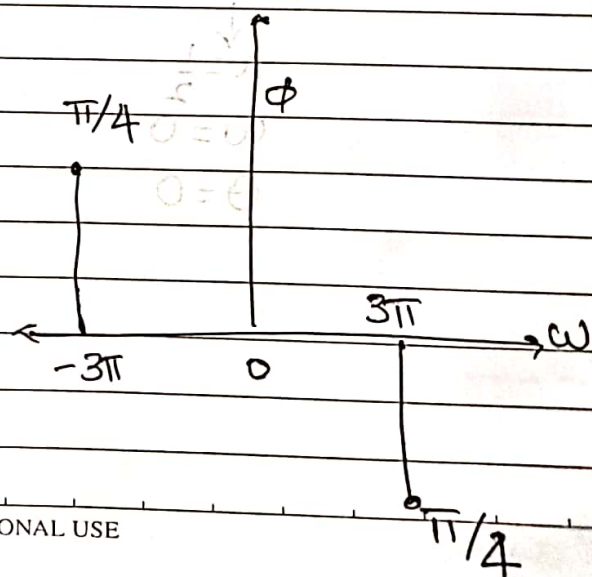
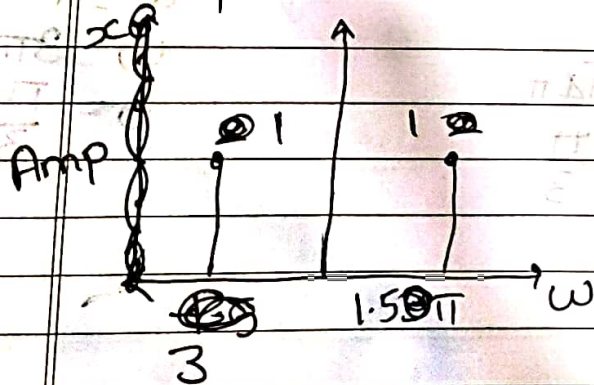
Ans: $a = 2, \omega = 3\pi, \theta = \frac{\pi}{4}$

Single Side
 1] Amplitude



Double Side

1] Amplitude



Q.2

$$x(t) = 2 \sin(10\pi t - \pi/3)$$

$$a=2, \omega=10\pi, \theta=\frac{\pi}{3}$$

$$Q. \sin^2[7\pi t - \pi/6] + \cos[3\pi t - \pi/3]$$

$$\frac{1 - \cos[14\pi t - \pi/3]}{2} + \cos[3\pi t - \pi/3]$$

$$= \frac{1}{2} \cos(0) - \frac{1}{2} \cos[14\pi t - \pi/3] + \cos[3\pi t - \pi/3]$$

$$\downarrow$$

$$a = \frac{1}{2}$$

$$\omega = 0$$

$$\theta = 0$$

$$\downarrow$$

$$a = -\frac{1}{2}$$

$$\omega = 14\pi$$

$$\theta = \frac{\pi}{3}$$

$$\downarrow$$

$$a = 1$$

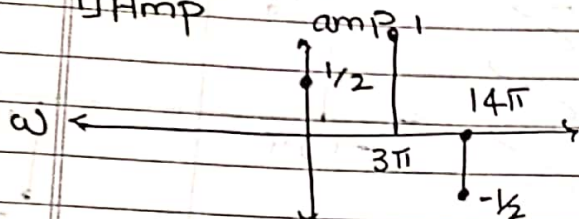
$$\omega = 3\pi$$

$$\theta = \frac{\pi}{3}$$

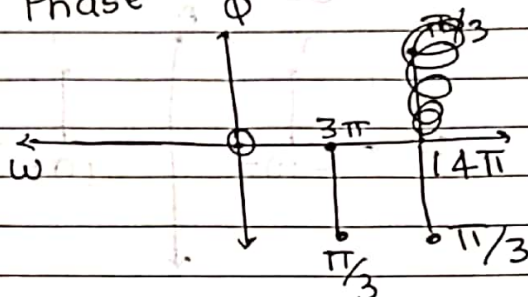
$e^{-j\omega t} = \cos(90^\circ - \theta) = \sin \theta$

Single Side :

Amplitude

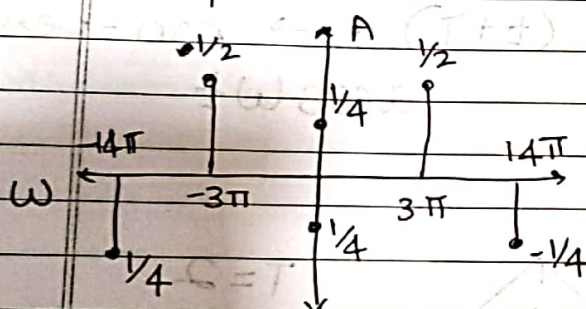


Phase ϕ

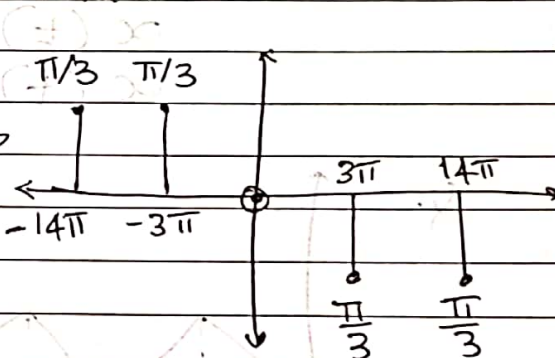


2] Double Sided

Amplitude



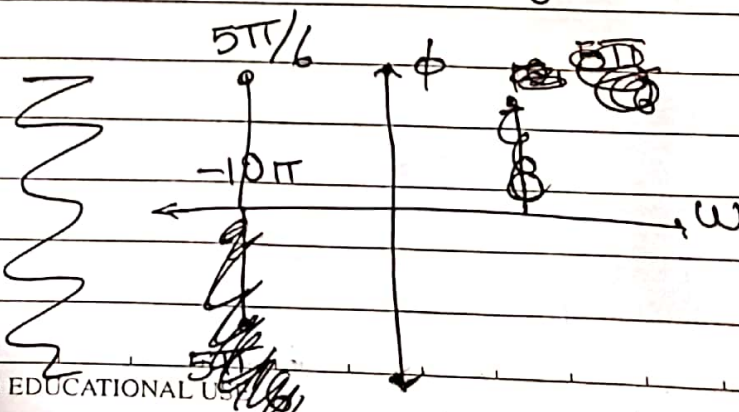
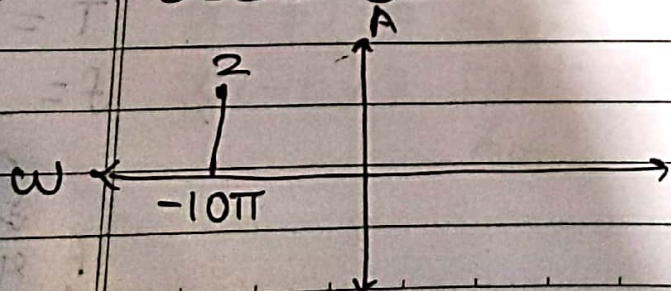
Phase



Q. $2 \sin(10\pi t - \pi/3)$
 $= 2 \cos(\pi/2 - (10\pi t - \pi/3))$
 $= 2 \cos(3\pi/2 - 10\pi t)$

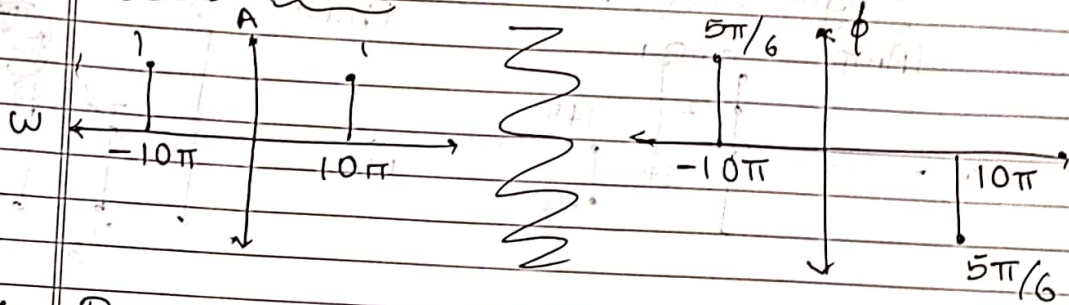
$a = 2, \omega = -10\pi, \theta = 5\pi/6$

Single Sided :



FOR EDUCATIONAL USE

Double Sided



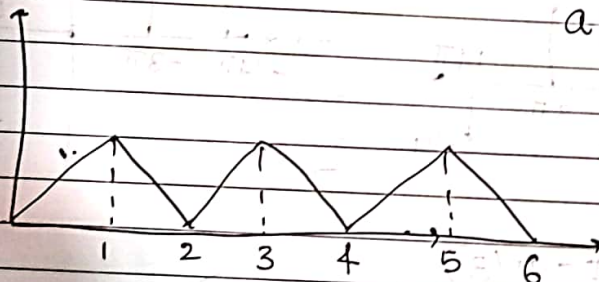
* Periodic-Non-Periodic Signal :

$$x(t) = x(t+T) \rightarrow \text{periodic}$$

$$x(t) \neq x(t+T) \rightarrow \text{non-periodic}$$

$$a \cos \omega t$$

Q.



$T=2$

$$x(t) = 2 \cos \frac{\pi}{4} t$$

$$\downarrow$$

$$a=2, \omega = \frac{\pi}{4}$$

$$2\pi f = \frac{\pi}{4}$$

$$\frac{2\pi}{T} = \frac{\pi}{4}$$

$$T=8$$

$$f = \frac{1}{8}$$

if rational
signal then
it's periodic

FOR EDUCATIONAL USE

FOR EDUCATIONAL USE

Sundaram

Q.

$$x(t) = 5 \cos 3\pi t + 7 \cos 6\pi t$$

↓

$$a = 5$$

$$\omega = 3\pi$$

$$2\pi f = 3\pi$$

$$f = \frac{3}{2}$$

↓

$$a = 7$$

$$\omega = 6\pi$$

$$2\pi f = 6\pi$$

$$f = 3$$

Both rational so overall signal is periodic

Q.

$$x(t) = \cos(10\pi t) + 5 \cos(2t)$$

↓

$$a = 1$$

$$\omega = 10\pi$$

$$2\pi f = 10\pi$$

$$f = 5$$

↓

periodic

↓

$$a = 5$$

$$\omega = 2$$

$$2\pi f = 2$$

$$f = \frac{1}{\pi}$$

↓

non-periodic

∴ non-periodic

Q.

Determining even & odd parts of signal

$$x(t) = \cos t + \sin t + \cos t \cdot \sin t$$

$$x(-t) = \cos t - \sin t - \cos t \cdot \sin t$$

even

odd

odd

Determine Required
Nyquist Sampling rate:

Q. $x(t) = 5 \sin 200\pi t$

i) $\omega_m = 200\pi$

$2\pi f_m = 200\pi$

$\therefore f_m = 100$

Now, $f_s = 2f_m = 2 \times 100 = 200 \text{ Hz}$

$\therefore$ min sampling rate = 200 Hz

ii) If $f_s = 100$, what is the discrete signal

$T = \frac{1}{f_s} = \frac{1}{100}$

$x(nT) = 5 \sin 200n\pi T$

$= 5 \sin \frac{200\pi n}{100}$

$= 5 \sin 2\pi n$

$= 0$

(iii) If $f_s = 300$

$T = \frac{1}{f_s} = \frac{1}{300}$

$x(nT) = 5 \sin 200\pi nT$

$= 5 \sin \frac{200\pi n}{300}$

$= 5 \sin \frac{2\pi n}{3}$

$\underline{\underline{3}}$

Q.

$$x(t) = 3 \cos 100\pi t$$

$$W_m = 100\pi$$

$$2\pi f_m = 100\pi$$

$$f_m = 50 \text{ Hz}$$

$$\therefore f_s = 2 \times f_m = \underline{\underline{100 \text{ Hz}}}$$

$$(ii) f_s = 200$$

$$T = \frac{1}{f_s} = \frac{1}{200}$$

$$\begin{aligned} x(nT) &= 3 \cos 100n\pi T \\ &= 3 \cos \frac{100n\pi}{200} \\ &= 3 \cos \frac{n\pi}{2} \end{aligned}$$

$$(iii) f_s = 75 \text{ Hz}$$

$$T = \frac{1}{75}$$

$$\begin{aligned} x(nT) &= 3 \cos \frac{100n\pi}{75} \\ &= 3 \cos \frac{20n\pi}{15} \end{aligned}$$